Tomato Leaf Disease Prediction Using CNN

Nirmitha S
M. Tech student
Department of Computer Science and Engineering
BNM Institution of Technology, Bangalore
nirmithas2002@gmail.com

S . Geetha
Professor
Department of Computer Science and Engineering
BNM Institution of Technology, Bangalore
geetha2016research@gmail.com

Chayadevi .M L
Professor
Department of Computer Science and Engineering
BNM Institution of Technology, Bangalore
hodcse@bnmit.in

Abstract— In many countries, tomatoes are regarded as an essential economic resource and are among the most valuable vegetables in the world. Nevertheless, these crops are susceptible to a number of illnesses that can reduce and damage the yield of healthy plants, thus it is crucial to identify these conditions early and accurately. Because of this, a more number of research has recently used deep learning (DL) models to automatically detect tomato leaf diseases. However, a few of these techniques just employ one DL framework, which necessitates a large amount of processing power to modify hyperparameters. The process of classification gets increasingly intricate this reliance. These networks also generate large dimensions, which further challenge classification. As a result, this study offers a technique that uses three compact convolutional neural networks (CNNs) to automatically detect tomato leaf illnesses. For a more succinct and sophisticated representation, it uses transfer learning to extract deep features from the final fully connected layer of the CNNs. In order to capitalize on the advantages of each CNN structure, it then integrates the features from the three CNNs. A comprehensive feature set with decreased dimensions is then selected and created using a hybrid feature selection method. Six distinct classifiers are used identification process.

Keywords— Real Time alert , CNN , Open CV ,Dib ,MEN,LEN, *Shape predictor,EARMAR.*

1.INTRODUCTION :

Farming has long been the main way for many nations in Africa and Asia to earn money. The full commercialization of farming has greatly affected our environment. Detecting plant diseases agriculture. Finding diseases early helps stop them from spreading to other plants, This may result in significant monetary losses. Plant illnesses significantly affect the agricultural sector This may result in anything from minor issues to the loss of entire farms.

Tomatoes rank among the most significant and widely eaten crops around the world. Current data indicates that global tomato production exceeds 180 million metric tons and generates about 8. 81 billion USD in exports. However, the production of tomatoes is on the decline due to the crop's vulnerability to numerous diseases. Tomato leaf diseases are a major cause of crop losses and financial hardship for farmers. The financial implications of agriculture are intimately related to the recognition of these leaf diseases. To maintain production and farmers' profitability, it is critical to promptly detect tomato leaf diseases and take the necessary action. Conventional disease detection techniques necessitate a manual inspection of afflicted leaves, either by visual inspection or chemical analysis. This can be time-consuming and may result in errors and unreliability because of human error. This situation is worsened by the lack of expertise among farmers and the scarcity of trained agricultural specialists capable of identifying diseases in remote areas, which hampers overall agricultural productivity. This kind of carelessness puts the world's food security at serious danger and costs those who grow tomatoes a lot of money. These issues could be successfully resolved and farmers could be assisted by using automated tools and methods for early disease identification.

Because of their accuracy and capacity for process automation, deep learning techniques—particularly Convolutional Neural Networks, or CNNs—have gained importance in the last several years for the detection of diseases in images. Accurate disease identification is made possible by CNNs' proficiency in identifying patterns and details in leaf images. In order to assure reliable and efficient disease detection, this project aims to develop a CNN-based model for recognizing illnesses in TLD using the deep learning techniques.

The proposed approach entails collecting and creating tomato leaves, model to recognize various diseases, and evaluating its efficacy in comparison to existing methods. By automating the identification process, this research offers farmers and other agricultural professionals a workable solution that enables them to act swiftly to minimize crop loss and boost production. It offers a versatile, discrete, and effective method of equipping new cars with driver monitoring devices, enhancing road safety.

2. LITERATURE REVIEW :

The Paper[1], discusses how plant diseases initially affect leaves before they spread to the rest of the plant, impacting both the type and quantity of crops, which significantly lowers agricultural output. To manage these diseases effectively, it is crucial to identify them accurately and early on. This research introduces a mode utilizes photos of 14 different healthy and sick plants, categorizing them into 38 classes from the Plant Village dataset. The model demonstrates potential for leaf diseases, achieving a training accuracy of 98. 01% and a test accuracy of 94. 33%. Future initiatives will involve collaborating with farms to gather various data, enhancing the CNN with additional architectures on more complex datasets, and developing a system for realtime disease monitoring.

According to the report [2], agriculture—which entails clearing land and growing crops for human consumption—is a vital sector of the Indian economy. However, illnesses that impede plant growth may be caused by different farming practices, seasonal variations, and climate variables. Plant protection depends on the prompt detection and diagnosis of these diseases. Standard classifiers including Backpropagation Neural Network (BPNN), Probabilistic Neural Network (PNN), Function of Rasial basis a neural network (RBFNN), Support Vector Machine (SVM), and KNearest Neighbors (KNN) have been utilized in the past for plant disease classification. However, these classifiers frequently struggle to strike a compromise between accuracy and speed. In order to classify plant diseases, this research presents an accurate and efficient image processing technique.

The suggested approach starts by using the backpropagation technique to ascertain the weights of neural network connections. The application of particle Swarm Optimization(PSO) enhances the neural network's connections in order to address problems like local optima and overfitting that are frequently encountered in conventional neural network training. Using photos of leaves afflicted with illnesses Spot, the tests 96.2%.

The Paper [3] According to the paper, a smartphone application was created to identify and categorize grape leaf illnesses [3]. To precisely locate and identify the impacted regions in the photos, this application makes use of the RCNN model for object identification, which is backed by an InceptionV2 backbone. It operates independently on mobile devices and has demonstrated a remarkable accuracy of 97. 9% based on tests conducted with images depicting grape diseases. By enabling early detection and management of diseases, the application assists farmers and cultivators in reducing crop losses and preventing disease spread.

The Paper[4] DLDPF is a framework presented in the study [4] that uses Keras, TensorFlow, and Python to integrate CNN utilizing cascade inception of AlexNet and GoogleNet. This system, which aims to automate the detection of diseases in apple plants, shows good performance compared to AlexNet, GoogLeNet, VGGNet16, and ResNet20. The framework, which is trained on an apple leaf dataset containing 10,888 training images and 2,801 test photos, has an accuracy rating of 97.62%. It requires 2.84 GB of memory and takes 33.45 minutes to train, which makes models. Investigating alternative models, like modified Recurrent Neural Networks (RNN), could be a promising research avenue for improving disease prediction in precision agriculture.

The paper [5] aim of plant pathology, as stated in paper [5], is to study diseases caused by pathogens and environmental factors.

This involves examining how to identify pathogens, understanding disease cycles, evaluating economic impacts, and exploring management strategies. Traditionally, disease diagnosis used spectroscopic technologies, which tend to be costly and require skilled operators. This research introduces a technique for identifying plant diseases with the help of a CNN. Initially, an image of a leaf sample is processed to remove healthy green pixels and separate color channels, allowing a focus on the damaged regions. The features from these affected areas are extracted and input into a CNN for disease classification. Once the illness is identified, solutions are sent to the user's mobile device via a GSM module.

Using the Plant Village dataset, the research [6] presents a Modified LeNet architecture which utilizes deep convolution neural networks (CNNs) to detect leaf diseases in maize. When trained to differentiate between, the CNN model attains an impressive accuracy of 97.89%. three disease categories and healthy maize leaves. This method, which is based on rigorous testing that changed variables including depth and kernel size, efficiently classifies many illnesses affecting maize leaves by focusing on both local and global aspects. The study emphasizes the significance of the suggested CNN design for enhancing agricultural information systems by showing that it is adaptable and effective in dealing with plant leaf diseases beyond those that impact maize.

Paper [7] CNNs, a deep learning technique, are utilized in the study [7] to tackle the primary threat that crop diseases represent to global food security. Using a set of 3852 photos from PlantVillage, a novel CNN model with 15 layers is developed to automatically identify crop diseases in maize. In addition to pictures of healthy plants, this collection includes pictures of three distinct diseases: Northern Corn Leaf Blight (NCLB), Common Rust (CR), and Gray Leaf Spot (GLS). The study also uses a technique called transfer learning to assess how well performs compared to several pretrained networks. proposed CNN model surpasses the leading pretrained network, DenseNet121, by 3.2 percent, even though it has fewer trainable parameters. This indicates that for identifying diseases in maize crops, the suggested CNN model outperforms other methods.

The study [8] looks at how the AlexNet model can be used to swiftly and precisely detect illnesses that harm maize leaves, which have a big influence on agricultural output and result in losses for businesses all over the world. With the assistance of the PlantVillage dataset, which contains diseases like Common Rust and leafspot ailments (Cercospora and Gray), the suggested deep neural network achieves a remarkable 99.16% accuracy rate. AlexNet and other convolutional neural networks automatically derive information from raw photos, which simplifies the illness identification process. In order to properly identify and treat these diseases—which is essential for safeguarding maize harvests and sustaining farmers' livelihoods—it is difficult to manage sizable datasets and optimize model iterations.

The study referenced as [9] investigates the enhancement of agricultural output by means of computer vision techniques that automate the detection of diseases affecting maize leaves. achieve this, three sophisticated designs of CNNs were assessed, employing strategies such as data augmentation, finetuning, and hyperparameter optimization via Bayesian methods. Both the training and testing processes utilized a strict fivefold crossvalidation method datasets, with the Plant Village dataset serving as the basis for evaluation. The findings indicated that all analyzed CNN models achieved impressive accuracy levels of up to 97%, highlighting the effectiveness of CNNs in disease classification when appropriate parameters are applied.

In order to overcome the shortcomings of traditional image processing techniques and enhance the precision of disease classification in maize plants, the work described in paper [10] investigates deep transfer learning techniques. It examines the performance of four specific deep learning models in identifying subtle illness indicators: Inception V3, ResNet50, InceptionResNetV2, and VGGNET. With a validation accuracy of 87. 51%, precisions of 90. 33%, and a recall rate of 99. 80%, ResNet50 stands out among these models. By using efficient and adaptable disease classification techniques designed for maize plants, these findings show the potential and benefits of deep transfer learning in raising agricultural output.

3.Methods:

A. A. The convolutional neural

3.1 CNN is a kind of deep neural network that is primarily utilized for visual image analysis. Convolutional neural networks, or CNNs, are a subclass of artificial neural networks used in deep learning for tasks including disease identification and picture classification. CNN successful in several fields. Among these are convolutional layers, which carry out the crucial function of feature extraction. A typical CNN structure consists of a number of distinct parts or layers. A CNN's conventional layout is depicted in Figure 1.

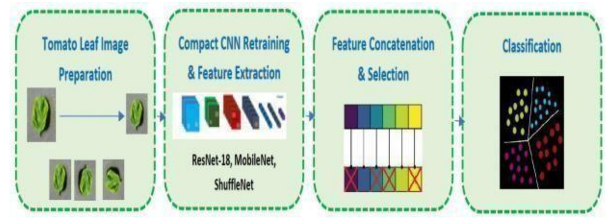

Fig.1 Depicting the classic architecture of CNN

Several crucial elements, referred to as layers in its architecture, make up a convolutional neural network:

- Convolutional Layer: This part employs a technique known as convolution to find characteristics like edges. This entails matching the input channels of the image with a filter or kernel of a particular size applied to the input volume. A feature map is the end product of this operation. We will produce k feature maps if we apply k filters.

- Pooling Layer: This part is in charge of downsampling. It minimizes the size of the representations, accelerates computations, and enhances the durability of some detected features. There are no parameters to learn for gradient

- descent in this layer. The hyperparameters related to pooling include filter size and stride, with a typical choice being a filter size of 2 X 2 and a stride of 2, which ultimately reduces the dimensions of the representations by a factor of 2.

- Fully Connected (FC) Layer: Fully connected layers make dense connections to the neurons in the layer below using the flattened output from the pooling levels. A softmax layer, which is frequently employed as the output layer for classification, can be connected to these neurons. The architecture of the network can influence the way many completely connected layers a CNN has.

3.2 DATA set:

Diseases in maize crops are identified using the Kaggle picture dataset of maize plants. It includes a variety of pictures showing various diseases of maize plants. To improve accuracy, the dataset is split into two sections: 30% testing and 70% training. The Harr cascade method employs a collection of attributes to recognize diseases in maize plant photos and to identify the photographs. This study utilized a dataset that was generated by augmenting the original TPD dataset.

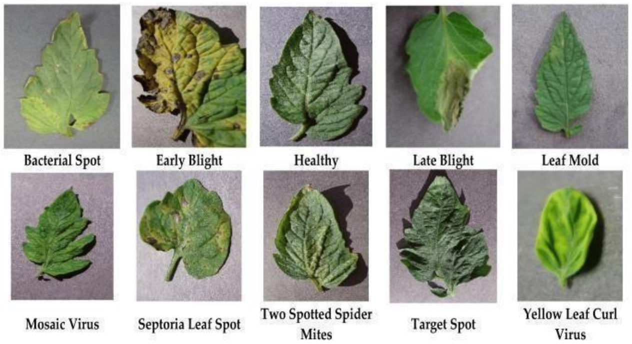

Fig 2: exemplary images for every class in the tomato dataset

This collection comprises approximately 2400 RGB images of various crop leaves—both healthy and unhealthy—affected by different diseases. The photographs have been used to create four different groups.

3.2: Proposed System

In order to ensure precise categorization and diagnosis, the process of identifying diseases in tomato plants is organized in a specific sequence. First, image data is obtained using Kaggle. During preparation, the images are resized to a uniform dimension of 224x224 pixels. Each disease category is assigned a distinct label, and the dataset is divided into training and testing sets. An initial feature extraction and disease identification are performed using a Haar Cascade classifier. After this, a Convolutional

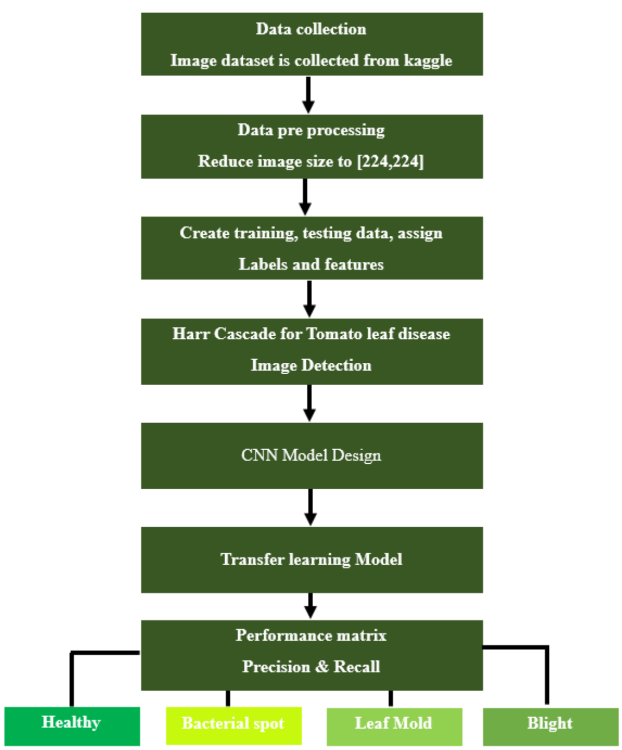

Fig 3. The steps of proposed system

- Data Gathering: Users provide image data of maize plants either through the mobile app or by uploading from their device's storage.
- Data Preparation: To guarantee uniform sizes, modify colors, and improve the dataset for model training, the stored data is preprocessed.
- Development of Models: The prepared dataset is used to train deep learning-based models through transfer learning methods. Models like VGG16, ResNet50, and ResNet100 are utilized for feature extraction.
- Model Refinement: To facilitate efficient use on mobile devices, the trained models are adjusted and converted into TensorFlow Lite (TFLite) format.
- Mobile App: The mobile application has features for taking photos, conducting realtime analysis with the trained models, showing the analysis results, and enabling user engagement.
- Image Recognition: Object detection algorithms identify images that show disease before analysis, guaranteeing precision in detecting maize plant diseases.
- System Oversight: System management tasks such as administration, monitoring, and maintenance are in place to support the system's reliability, security, and performance.

3.3 The Proposed CNN

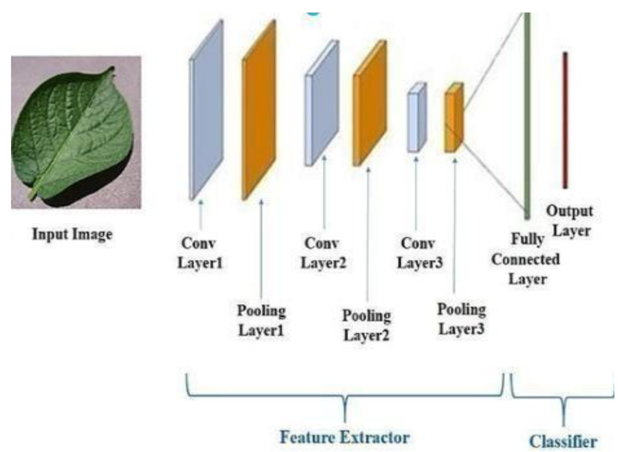

Fig 4. Architecture of the CNN experiment

This model consists of two convolutional layers, each containing 32 filters of size 3x3. Following that, there is a max pooling layer with a 2x2 window. The dimensions of the input are set to 256 × 256 x 3. The following block contains two convolutional layers, each with 64 filters of size 3x3, along with a max pooling layer that replicates the configuration of the first block.

The configuration of the first block is duplicated in the third block, which contains 128 convolutional filters and two convolution layers, along with a max pooling layer. After the fully connected layer, there is a dropout layer with a rate of 0.5. Die Ausgabeschicht besteht aus 38 Einheiten, was der Anzahl der Klassen entspricht. Except for the output layer that utilizes a softmax function, all layers employ the Rectified Linear Unit (ReLU) as their activation function. In this experiment, the settings used were the Adam optimizer, a batch size of 30, a dropout rate of 0.5, and a total of 10 epochs.

E. Metrics for assessing the model

To assess the effectiveness of the model, three metrics were utilized in these experiments: True Positive (TP), True Negative (TN), False Positive (FP), and False Negative (FN).

$$\text{True Positive Rate} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

$$\text{True Negative Rate} = \frac{\text{TN}}{\text{TN} + \text{FP}}$$

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

Fig 5.Measuring Models

The metric for evaluating the performance of trained models. It offers a thorough examination of the model's predictions and how accurate they are in comparison to the real classes. When it comes to binary classification tasks, like distinguishing between several illnesses in maize plants, the confusion matrix is particularly useful.

	Actually Positive (1)	Actually Negative (0)
Predicted Positive (1)	True Positives (TPs)	False Positives (FPs)
Predicted Negative (0)	False Negatives (FNs)	True Negatives (TNs)

Fig 6. Confusion matrix

To assess the model's effectiveness in detecting illnesses on tomato leaves, a confusion matrix is employed that demonstrates how proficiently the model classifies the leaves. Four key elements exist to this matrix: True Positives (TPs), which are the diseased leaves that the model classified correctly; True Negatives (TNs), indicating the healthy leaves that were correctly identified; False Positives (FPs), where healthy leaves are wrongly classified as having disease; and False Negatives (FNs), where diseased leaves are mistakenly categorized as healthy. Grasping these values is essential for calculating important evaluation metrics score, which contribute to a reliable assessment of the model's performance.

IV. RESULTS AND DISCUSSION

demonstrates the model's excellent accuracy and dependability as it reaches and stabilizes at approximately 97% by the 10th epoch. The sophisticated design of ResNet101, which successfully discerns complex patterns and features in maize plant images and utilizes them for precise disease classification, contributes to this outstanding performance. This result highlights the model's potential for practical applications in agriculture, allowing for rapid and precise disease identification to enhance crop management. The results are illustrated in Fig 8 and Table 2.

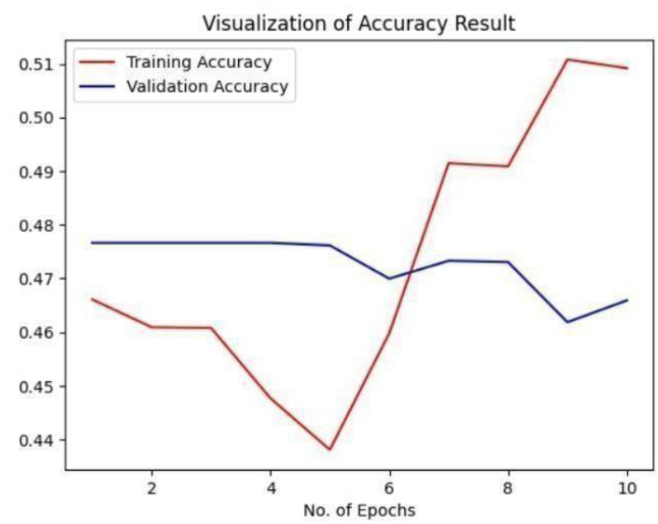

Fig 7. Train & validation accuracy obtained using rennet[10]

	precision	recall	f1-score	support
0	1.00	0.98	0.99	100
1	0.83	1.00	0.90	100
2	0.95	1.00	0.98	100
3	1.00	0.76	0.86	100
accuracy			0.94	400

Fig 8. Analysis Result of ResNet101 model

Deep Learning Models	No of Epochs	Accuracy
Resnet 100	10	0.97
Resnet 50	10	0.69
VGG 16	10	0.94

TABLE 1. Performance of DL models

Fig 8 . Web interface

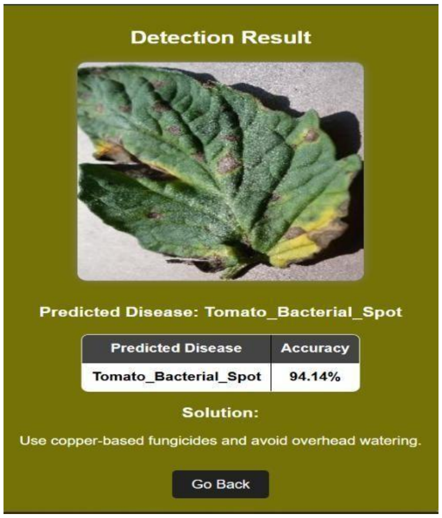

Fig 9 . Disease detection

CONCLUSION

Classifying Tomato leaf disease pridection using deep learning. By combining several CNN architectures-MobileNetV2, VGG16, and ResNet50- our model effectively extracts high-quality features while lowering computational requirements. The Streamlining the dataset and choosing the most pertinent features for classification.

The experimental results indicate that the proposed framework attains a remarkable classification accuracy exceeding 97%, and it surpasses conventional single-CNN models regarding both accuracy and efficiency. Utilizing six different classifiers results in a more reliable detection method, ensuring improved generalization across various illness categories.

This research contributes, in practical terms, to the development of sophisticated agricultural technologies that assist farmers in early disease identification. greatly improve real-time case monitoring and decision-making in smart farming systems or mobile applications.

Future research can investigate broadening this approach to encompass a wider variety of crop diseases optimizing lightweight CNN architectures for immediate deployment, and integrating real-world environmental factors into the dataset to enhance model robustness, Moreover, further advancements in feature selection methods and classifier optimization can refine accuracy and efficiency even more feature selection methods and classifier optimization can refine accuracy and efficiency even more
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